

1. For which positive integers k is the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}$ convergent?

We know if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, $\sum a_n$ is convergent

$$a_n = \frac{(n!)^2}{(kn)!}$$

$$a_{n+1} = \frac{[(n+1)!]^2}{(k(n+1))!}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{[(n+1)!]^2}{(k(n+1))!}}{\frac{(n!)^2}{(kn)!}} \right|$$

$$= \left| \frac{[(n+1)!]^2 (kn)!}{(k(n+1))! (n!)^2} \right|$$

$$= \left| \frac{(n+1)^2}{(kn+k)(kn+k-1)\cdots(kn+1)} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{(kn+k)(kn+k-1)\cdots(kn+1)} \right|$$

$$\approx \lim_{n \rightarrow \infty} \left| \frac{n^2}{(kn)^k} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{1}{k^k n^{k-2}} \right|$$

Since $n \rightarrow \infty$, $n^{k-2} \rightarrow \infty$, we only care about if this limit less than 1, then we choose to ignore n^{k-2}

we can get $\left| \frac{1}{k^k} \right| < 1$, since k is positive integer, $k > 1$

In conclusion, $k > 1$